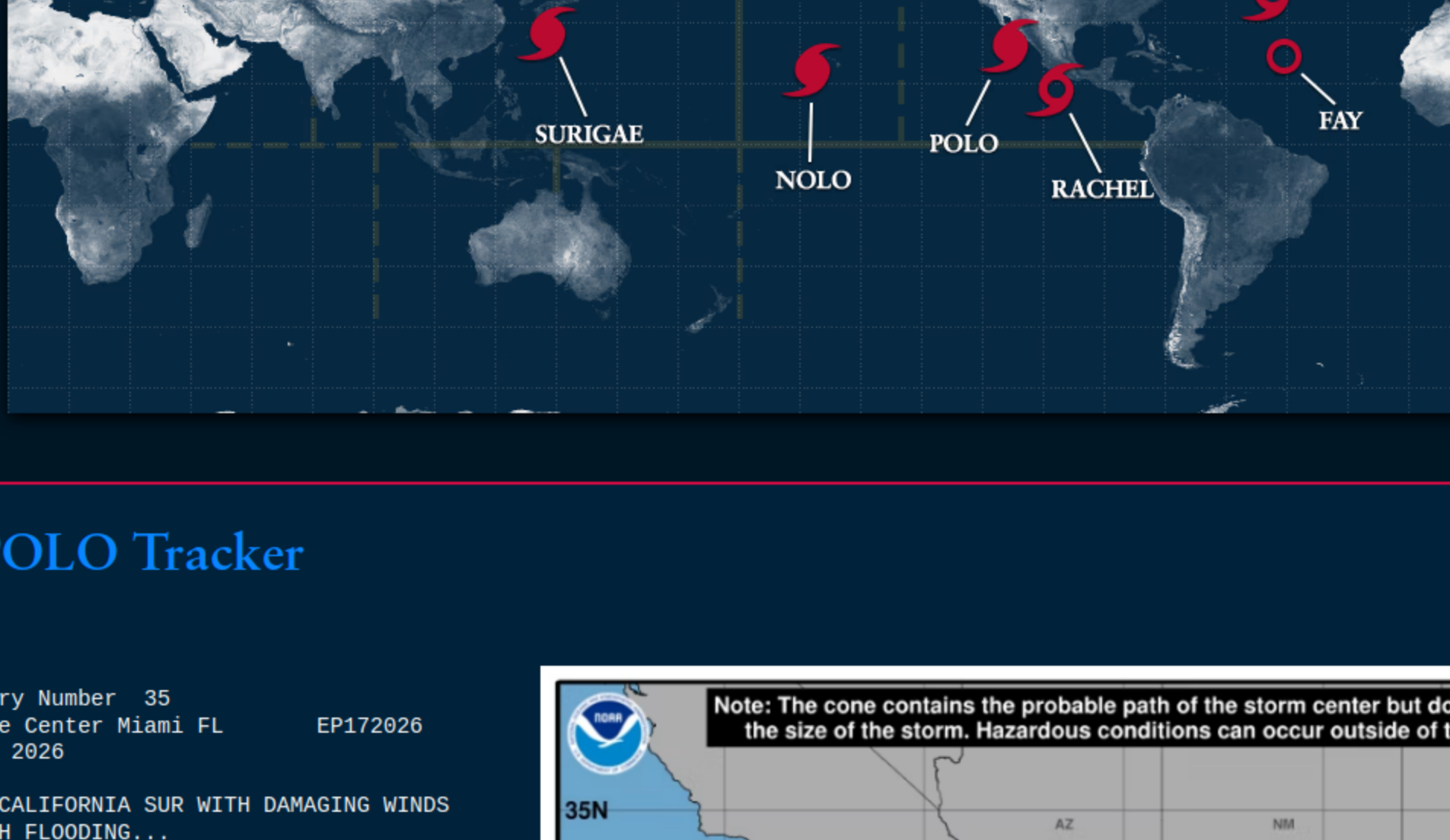


STORMDOM

Global Storm Tracker

Home ▾ Indian Ocean ▾ West Pacific ▾ South Pacific ▾ Central Pacific ▾ East Pacific ▾ Atlantic



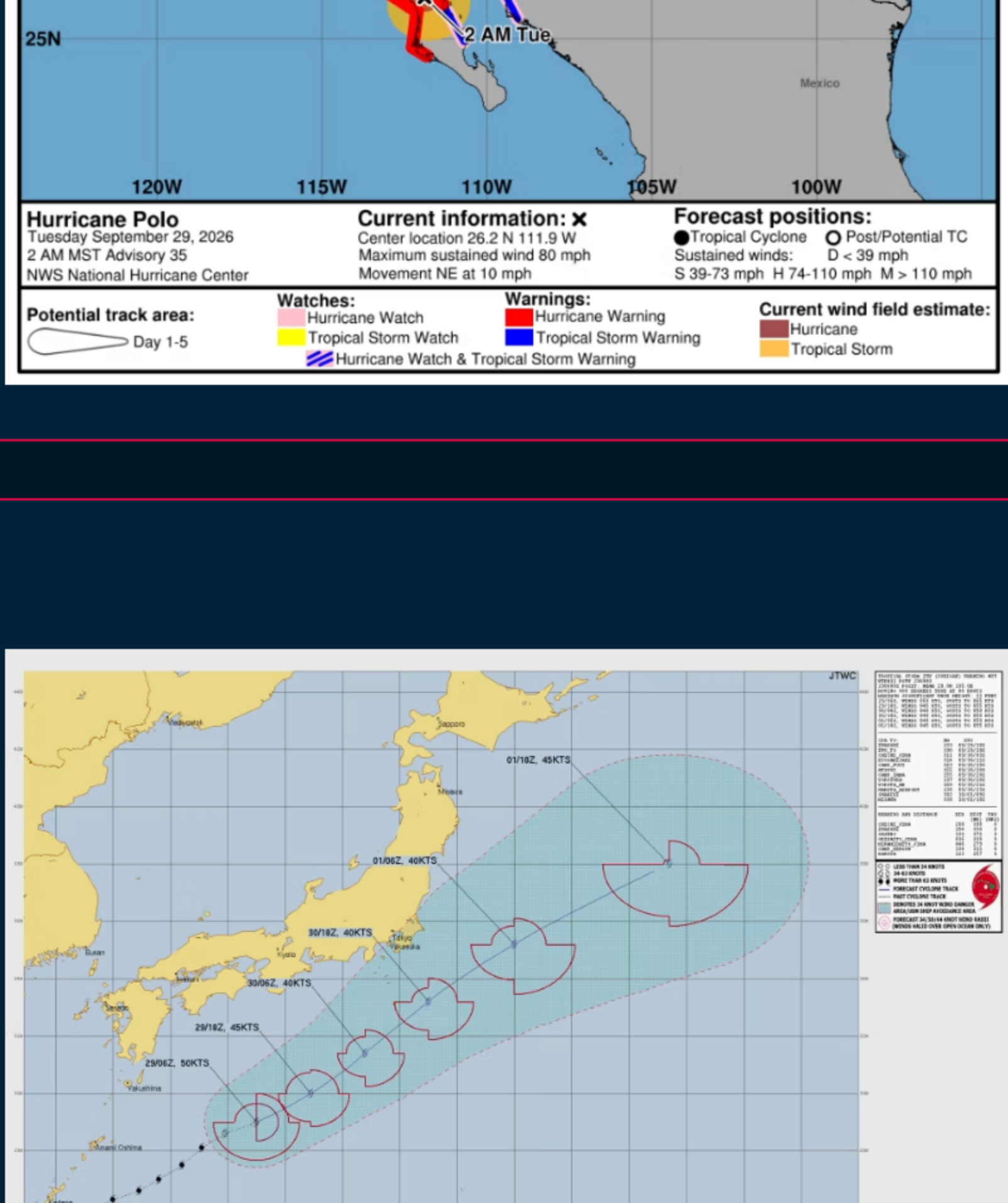
Hurricane POLO Tracker

Hurricane Polo Advisory Number 35
NWS National Hurricane Center Miami FL
200 AM MST Tue Sep 29 2026

...POLO LASHING BAJA CALIFORNIA SUR WITH DAMAGING WINDS
LIFE-THREATENING FLASH FLOODING...
...CONDITIONS EXPECTED TO DETERIORATE IN SONORA EARLY T
MORNING...

SUMMARY OF 200 AM MST...0900 UTC...INFORMATION

LOCATION...28.2N 111.9W
ABOUT 35 MI...55 KM NNM OF LORETO MEXICO
ABOUT 140 MI...225 KM SSW OF GUAYMAS MEXICO
MAXIMUM SUSTAINED WINDS...80 MPH...130 KM/H
PRESENT MOVEMENT...NE OR 40 DEGREES AT 10 MPH...17 KM/H
MINIMUM CENTRAL PRESSURE...978 MB...28.88 INCHES

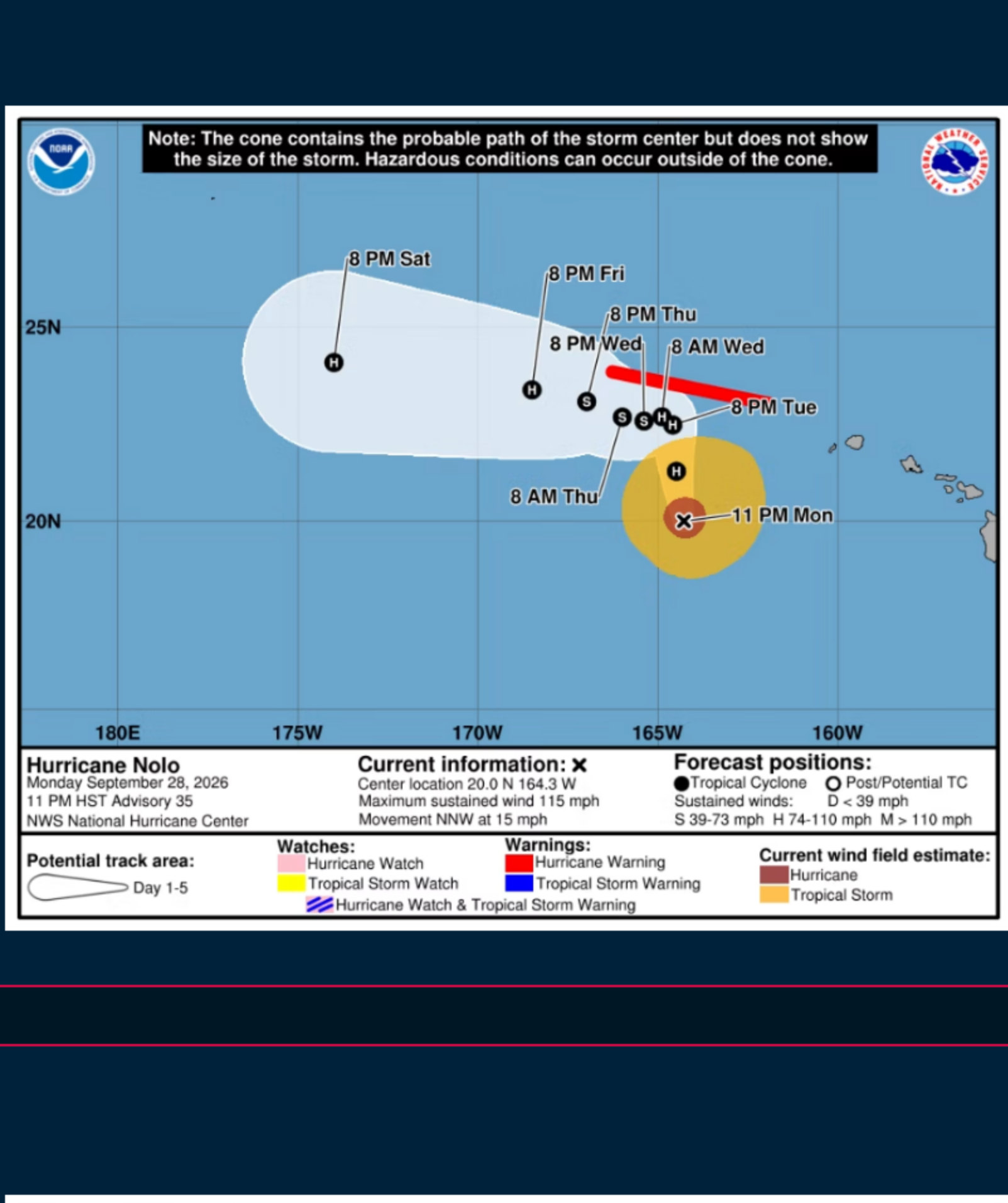


Typhoon SURIGAE Tracker

1. TROPICAL STORM 25W (SURIGAE) WARNING NR 025
01 ACTIVE TROPICAL CYCLONE 1N NORTHWESTPAC
Issued by NWS National Hurricane Center Honolulu HI
WIND RADII VALID OVER OPEN WATER ONLY

WARNING POSITION:
290600Z --- NEAR 29.0N 135.0E
MOVEMENT PAST SIX HOURS - 065 DEGREES AT 10 KTS
POSITION ACCURATE TO WITHIN 030 NM
POSITION BASED ON CENTER LOCATED BY A COMBINATION
SATELLITE AND RADAR

PRESENT WIND DISTRIBUTION:
MAX SUSTAINED WINDS - 050 KT, GUSTS 065 KT
WIND RADII VALID OVER OPEN WATER ONLY
RADIUS OF 050 KT WINDS - 040 NM NORTHEAST QUADRANT
040 NM SOUTHEAST QUADRANT
090 NM SOUTHWEST QUADRANT
090 NM NORTHWEST QUADRANT
RADIUS OF 034 KT WINDS - 060 NM NORTHEAST QUADRANT
060 NM SOUTHEAST QUADRANT
080 NM SOUTHWEST QUADRANT
080 NM NORTHWEST QUADRANT
REPEAT POSIT: 29.0N 135.0E



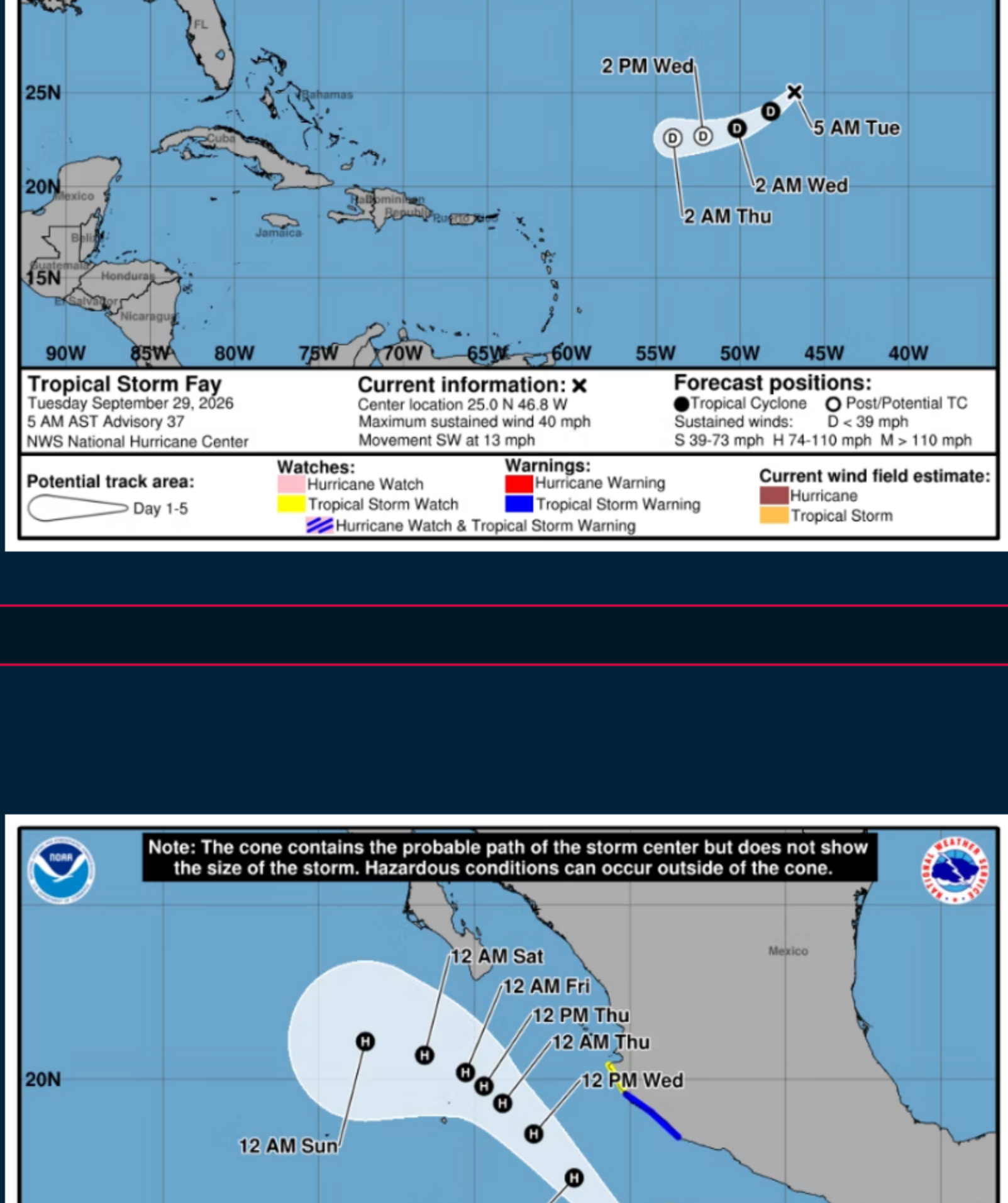
Hurricane NOLO Tracker

Hurricane Nolo Advisory Number 35
NWS Central Pacific Hurricane Center Honolulu HI
Issued by NWS National Hurricane Center Miami FL
1300 PM HST Mon Sep 28 2026

...CENTER OF MAJOR HURRICANE NOLO ENTERS THE PAPAHAUUM
MARINE NATIONAL MONUMENT...
...DANGEROUS CONDITIONS SPREADING ACROSS THE SOUTHEAST
OF THE MONUMENT...

SUMMARY OF 1100 PM HST...0900 UTC...INFORMATION

LOCATION...28.0N 164.2W
ABOUT 380 MI...480 KM SSE OF FRENCH FRIGATE SHOALS
ABOUT 425 MI...680 KM W OF HONOLULU HAWAII
MAXIMUM SUSTAINED WINDS...115 MPH...185 KM/H
PRESENT MOVEMENT...WNW OR 330 DEGREES AT 15 MPH...24 KM
MINIMUM CENTRAL PRESSURE...958 MB...28.29 INCHES



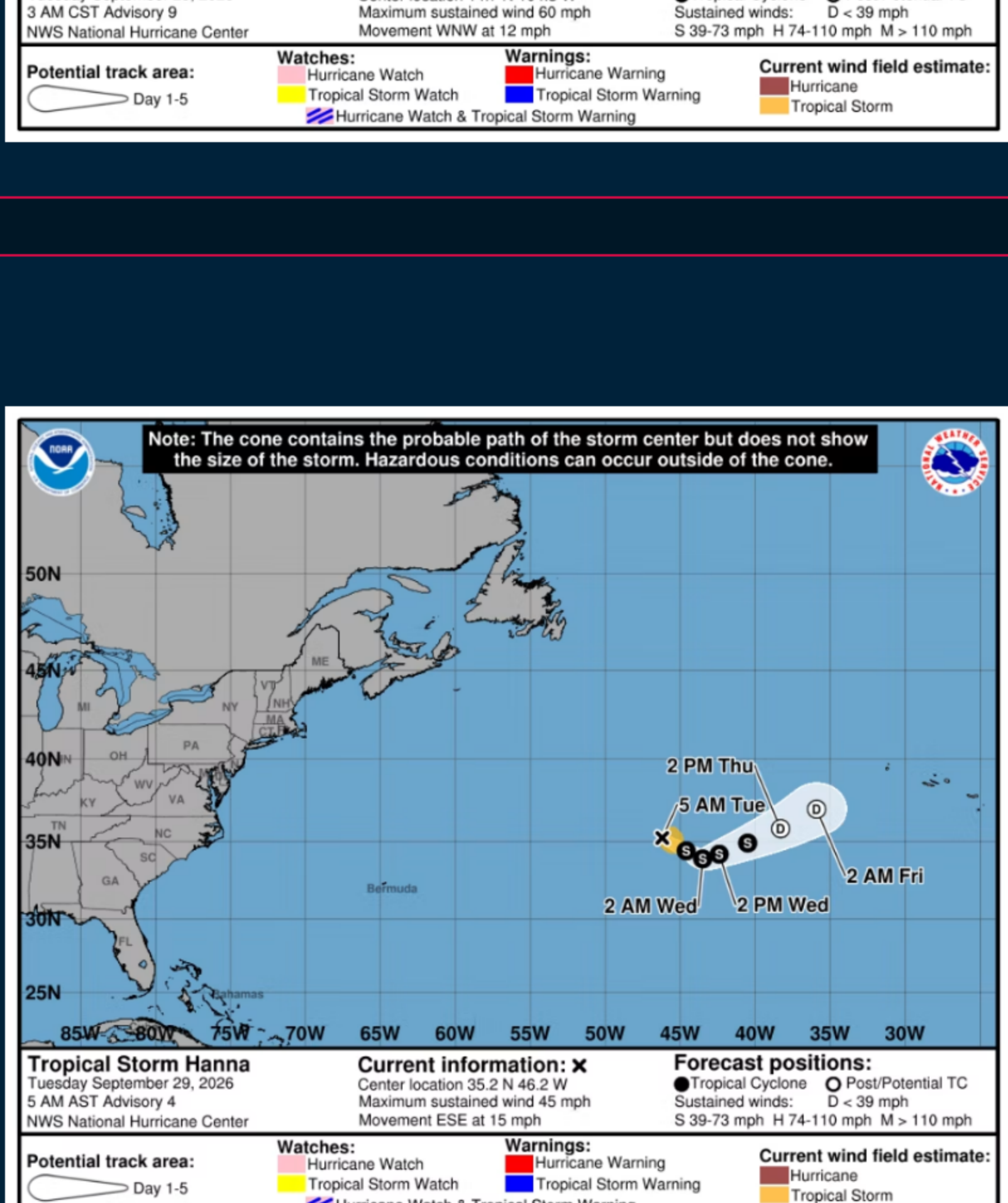
Tropical Depression FAY Tracker

Tropical Storm Fay Advisory Number 37
NWS National Hurricane Center Miami FL
300 AM AST Tue Sep 29 2026

...FAY CONTINUES TO MOVE SOUTHWESTWARD OVER THE ATLANTI

SUMMARY OF 300 AM AST...0900 UTC...INFORMATION

LOCATION...25.0N 46.8W
ABOUT 1465 MI...2360 KM WSW OF THE AZORES
MAXIMUM SUSTAINED WINDS...40 MPH...65 KM/H
PRESENT MOVEMENT...SW OR 235 DEGREES AT 12 MPH...20 KM/H
MINIMUM CENTRAL PRESSURE...1008 MB...29.77 INCHES



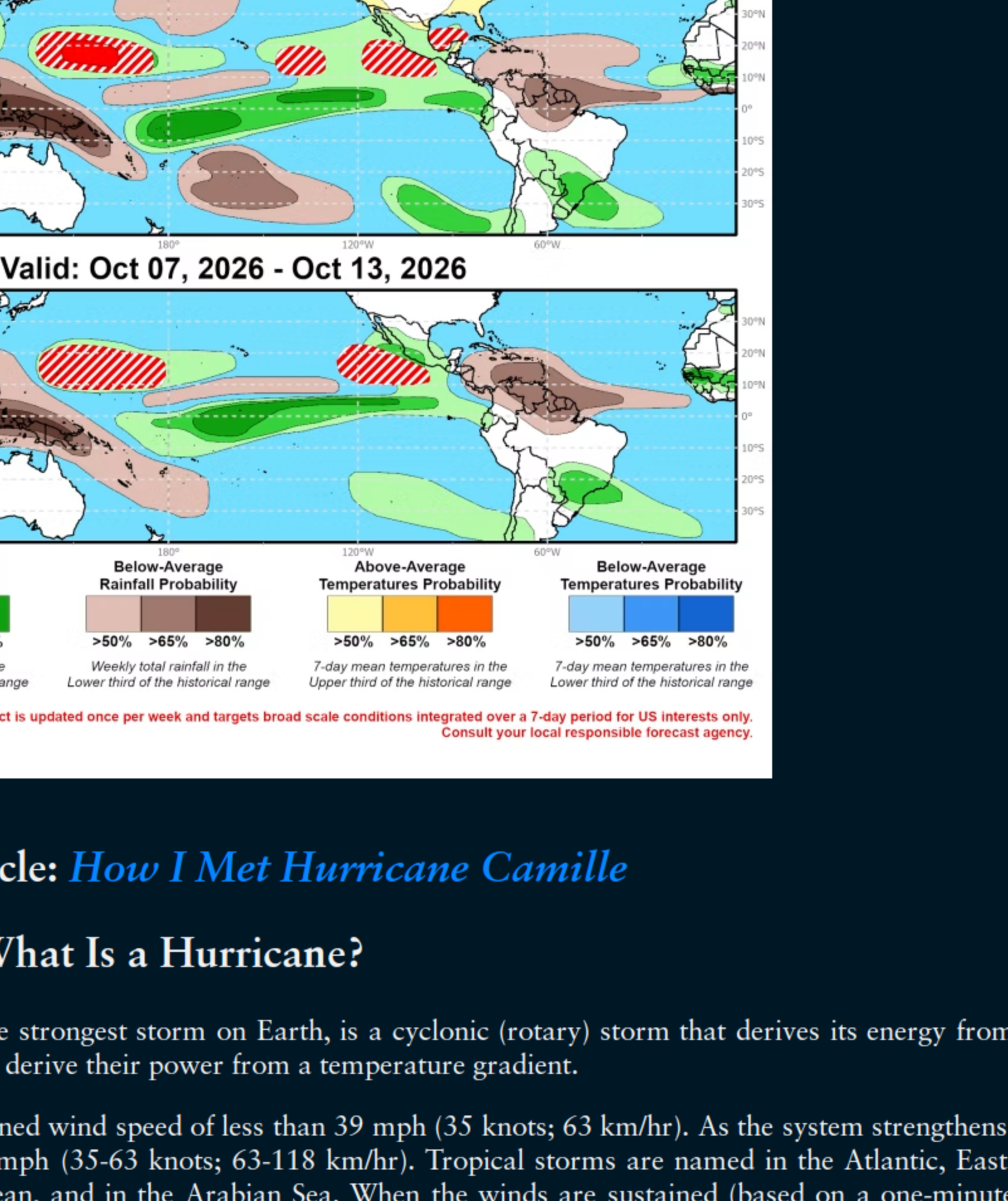
Tropical Storm RACHEL Tracker

Tropical Storm Rachel Advisory Number 9
NWS National Hurricane Center Miami FL
500 AM CST Tue Sep 29 2026

...RACHEL EXPECTED TO BECOME A HURRICANE DURING THE NEX
...

SUMMARY OF 300 AM CST...0900 UTC...INFORMATION

LOCATION...14.7N 104.3W
ABOUT 265 MI...430 KM SSW OF LAZARO CARDENAS MEXICO
ABOUT 495 MI...690 KM SSE OF CABO CORRIENTES MEXICO
MAXIMUM SUSTAINED WINDS...60 MPH...95 KM/H
PRESENT MOVEMENT...WNW OR 300 DEGREES AT 12 MPH...19 KM
MINIMUM CENTRAL PRESSURE...992 MB...29.30 INCHES



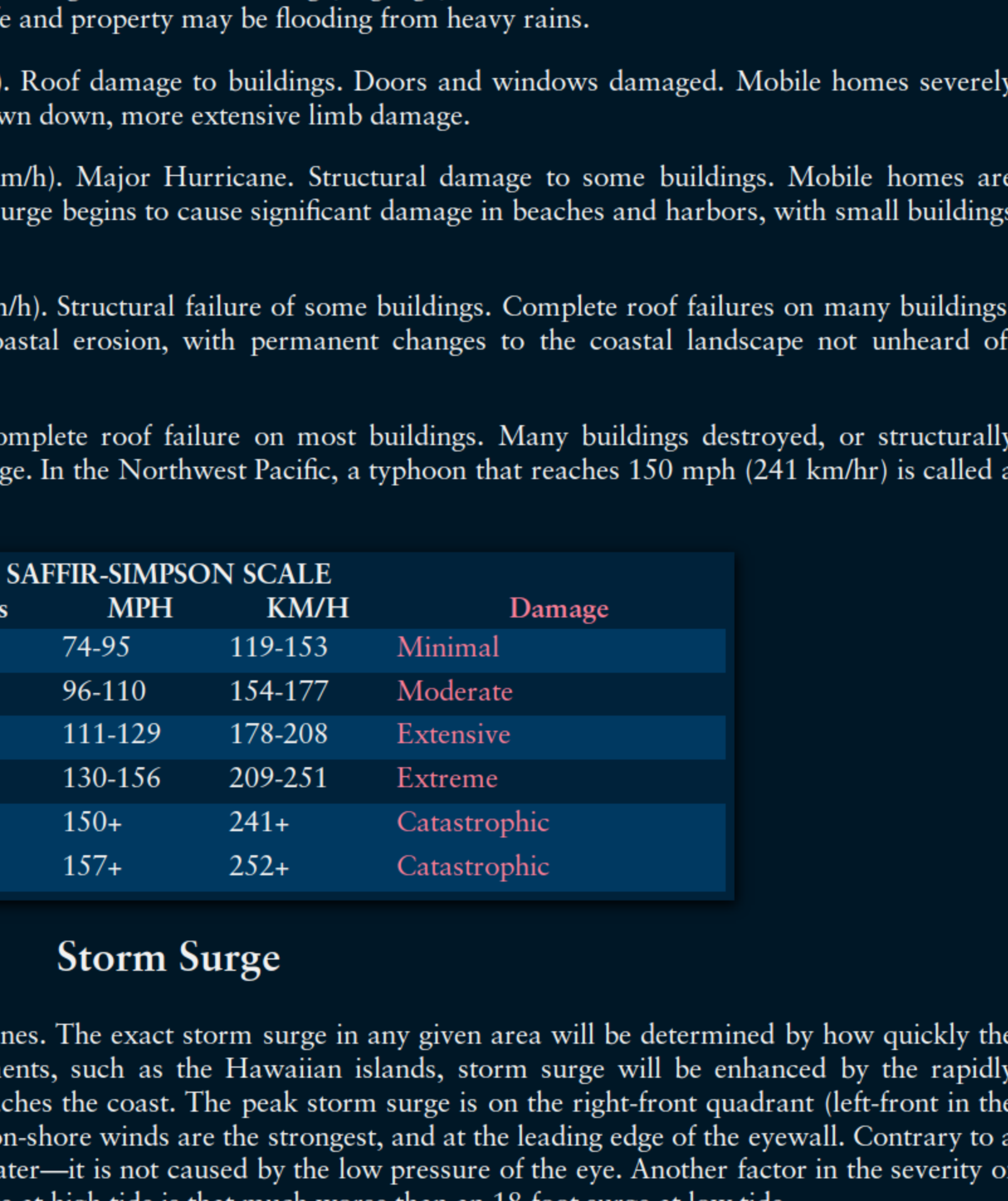
Tropical Storm HANNA Tracker

Tropical Storm Hanna Advisory Number 4
NWS National Hurricane Center Miami FL
500 AM AST Tue Sep 29 2026

...HANNA HOLDS STEADY WHILE MAINTAINING AN EAST-SOUTHEA

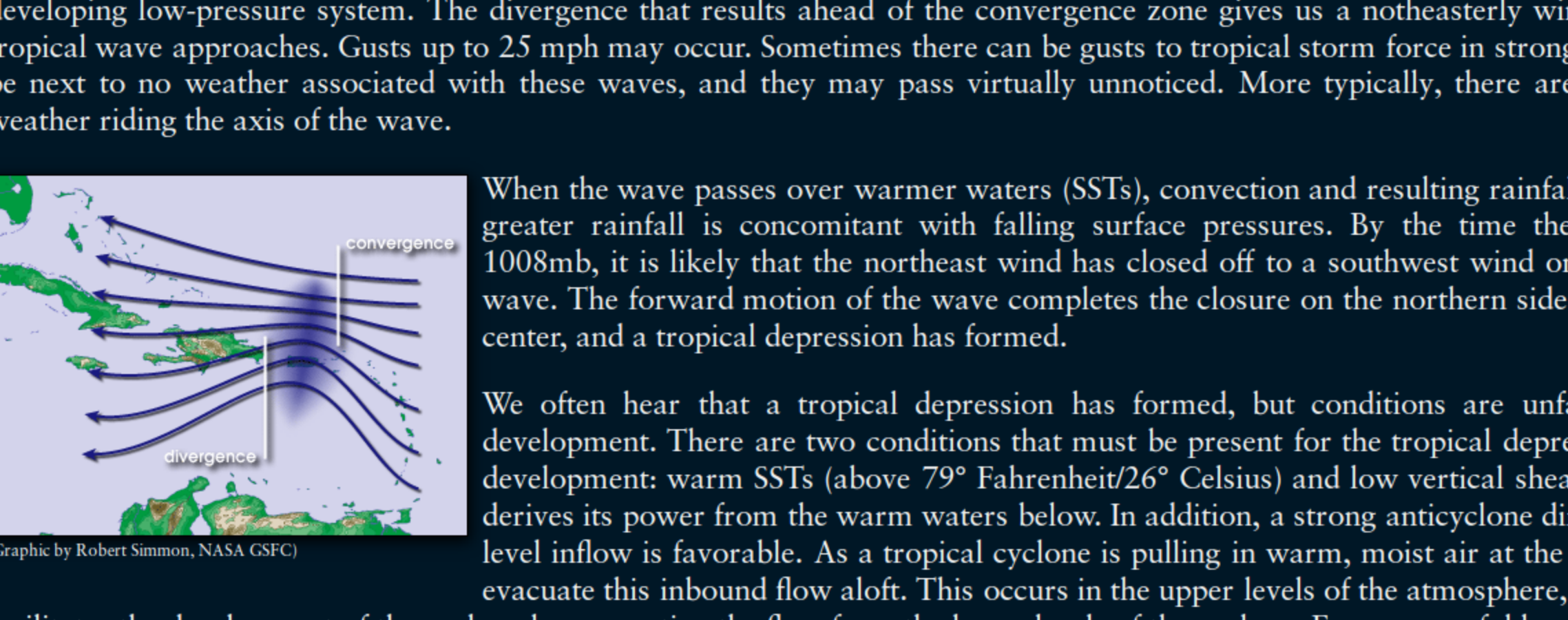
SUMMARY OF 500 AM AST...0900 UTC...INFORMATION

LOCATION...35.2N 46.2W
ABOUT 1885 MI...3045 KM ENE OF BERMUDA
ABOUT 1880 MI...1740 KM W OF THE AZORES
MAXIMUM SUSTAINED WINDS...45 MPH...75 KM/H
PRESENT MOVEMENT...ESE OR 115 DEGREES AT 15 MPH...24 KM
MINIMUM CENTRAL PRESSURE...1002 MB...29.59 INCHES



Global Tropics Hazards Outlook

Climate Prediction Center
Week 2 - Valid: Sep 30, 2026 - Oct 06, 2026
Week 3 - Valid: Oct 07, 2026 - Oct 13, 2026



Issued: 09/22/2026
Forecaster: Novella
Graphic provided by Climate Prediction Center

See my new article: [How I Met Hurricane Camille](#)

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature gradient.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (35 knots; 63 km/hr). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (35-63 knots; 63-118 km/hr). Tropical storms are named in the Atlantic, East, Central and North Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots; 119 km/hr), the storm becomes: In the Atlantic Ocean, East Pacific, Central Pacific (east of the International Dateline) and Southeast Pacific (east of 160°E) a Hurricane; in the Northwest Pacific (west of the International Dateline) a Typhoon; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a Severe Tropical Cyclone; in the North Indian Ocean a Severe Cyclonic Storm; and in the South Indian Ocean (west of 90°E) a Tropical Cyclone.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph; 119-153 km/h). Damage is limited to foliage, signage, boats and mobile homes. There is no significant damage to buildings. The main threat to life and property may be flooding from heavy rains.

Category 2 – 83-95 knots (96-110 mph; 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers damaged by storm surge. Some trees blown down, more extensive limb damage.

Category 3 – 96-112 knots (111-129 mph; 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph; 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extreme storm surge damage and flooding. Severe coastal erosion, with permanent changes to the coastal landscape not unheard of. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph; 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/hr) is called a Super Typhoon.

Category	SAFFIR-SIMPSON SCALE			Damage
	Knots	MPH	KM/H	
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	130+	150+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly decreasing ocean depth as the wind-driven surge approaches the coast. The peak storm surge is on the right-front quadrant (left-front in the Southern Hemisphere) of the eyewall at landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular myth, the storm surge is entirely wind-driven water—it is not caused by the low pressure of the eye. Another factor in the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

By Jonathan Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation after about the middle of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a “convergence” wave, where lines of equal atmospheric pressure are pressed together between the high-pressure ridge to the north and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a northeasterly wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There can be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather riding the axis of the wave.

(Graphic by Robert Simons, NASA GSFC)

When the wave passes over warmer waters (SSTs), convection and resulting rainfall are enhanced. This greater rainfall is concomitant with falling surface pressures. By the time these pressures fall to 1008mb, it is likely that the wavehead wind has closed off to a southwest wind on the broadside of the wave. The forward motion of the wave completes the closure on the northern side of a broad low-level center, and a tropical depression has formed.

We often hear that a tropical depression has formed, but conditions are unfavorable for further development. There are two conditions that must be present for the tropical depression to continue its development: warm SSTs (above 79° Fahrenheit/26° Celsius) and low vertical shear. A tropical cyclone derives its power from the warm waters below. In addition, a strong anticyclone directly above the low-level inflow is favorable. As a tropical cyclone is pulling in warm, moist air at the surface, it must also evacuate this inbound flow aloft. This occurs in the upper levels of the atmosphere, where high pressure facilitates the development of the cyclone by evacuating the flow from the lower levels of the cyclone. Every powerful hurricane has an equally powerful high pressure system over it. At the surface, the air spirals inward in a counter-clockwise direction, rises in the developing center, and spirals out at the top in a clockwise direction. In the Southern Hemisphere, it is reversed: clockwise inbound, counterclockwise outbound.

If the upper-level high pressure system does not remain over the cyclone, it means there is shear ahead. This is a strong jet of air that is blowing directly over the cyclone, ripping the cloud tops off the deep convection. This has the effect of breaking down the whole mechanism. This is known as vertical shear. Vertical shear usually comes from a westerly or southerly direction, and can occur if the cyclone is located in an unfavorable subsidence near a cold front or upper-level low pressure system. Another factor that can interfere with the development of a tropical cyclone is position. A subsidence is the sinking of air. This happens on the edge of an upper-level high-pressure system. Subsiding air has the effect of suppressing thunderstorm formation. This also is why a tropical storm that tries to form near an established hurricane has a very difficult time—the cyclone is on the edge of the hurricane's upper-level outflow, and may have to contend with both subsidence and shear. The effect of shear on a cyclone can range from a failure to develop further to catastrophic collapse of the tropical cyclone's support structure.

Tropical cyclones have the low-level circulation and the upper-level circulation (outflow), whose formation was discussed above. There is also a mid-level circulation. The mid-level circulation is similar in structure to the low-level circulation, and is critical to the survival of a tropical cyclone that is passing over land. The low-level circulation can be severely disrupted, or even dissipated, by interaction with land, especially mountainous terrain. If the mid-level circulation remains intact, the cyclone can regenerate rapidly when it reemerges over water, providing other factors are favorable.

(Image courtesy NOAA)

Assuming—all the ingredients are in place—warm SSTs, upper-level high pressure, and falling surface pressures—the cyclone will develop and reach a point of rapid intensification. It is one of nature's perfect machines. As warm waters feed the convection swirling around the center, heavy rainfall lowers surface pressures, high pressure aloft evacuates the inflow, which intensifies the inflow of warm, moist air, which in turn increases the rainfall and brings about a more rapid fall in central pressure.

Eye formation begins when a tropical storm reaches approximately 65mph, provided conditions are favorable for strengthening to continue. The eyewall begins to make its appearance, usually on the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and the eastern (Northern Hemisphere) edge of the center. As the system becomes better organized and the eastern (Northern Hemisphere) edge of the center.

smaller, the center contracts from about 200 miles across to roughly 90 miles at this stage. An increase in rotational velocity accompanies the stronger, more defined center. The inflow is spiralling in ever faster as it is evacuated up through the developing eyewall and out by the high pressure outer flow structure. The eye begins to appear as a clear spot in the center, which is a sinking of the air. The eyewall creates subsidence that not only helps clear the eye, but can also produce a feature known as the “moat”, the air here is an area of relatively weak convection outside the core of the tropical cyclone. The combination of eye and moat can make the storm's core look like a doughnut. The eye can have a diameter anywhere from 10 to 40 miles across.

Tropical cyclones can exhibit a great deal of durability provided that the upper level support remains and the southerly (Northern Hemisphere), or tropical, inflow is present. Here are some examples: In 1998, Hurricane Mitch developed into a supermassive Category 5 hurricane, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts well exceeding 200 mph—which roared up the Mississippi Valley and eventually exited off the East Coast. Camille maintained tropical storm strength as far as Memphis, Tennessee. Most hurricanes will diminish in strength rapidly after landfall, reaching tropical depression strength by 48 to 72 hours. The main threat from the dying storm is from tornadoes and inland flooding. The right-hand quadrant of a hurricane or typhoon (in the Northern Hemisphere, left-hand in the Southern Hemisphere) is most frequently the strong side of the storm. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the right-hand side of the hurricane contains the strong on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life, will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and finally Landfall.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area 200 to 300 miles across. The result is weakening. It is possible for a hurricane to stall in one area long enough that it dissipates. In 2004, Hurricane Frances stalled off the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 1 hurricane, instead of the earlier expected Category 3.

Entraining dry air. Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as part of their interaction with frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains deeply enough, the result can be dissipation.

Moving over cool water. Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begins to weaken. Eventually this causes dissipation, particularly in the East Pacific. In the Atlantic, if the storm is caught in the mid-latitude westerlies and begins to recurve to the northeast, it generally becomes an extratropical storm by the time it has reached about 45° W. This is the so-called “graveyard” of Atlantic hurricanes. The storm becomes extratropical when it has transitioned to a cold-core, baroclinically driven system, and eventually becomes a gale in the north Atlantic, or is absorbed by a large gale. There have been times when a hurricane passes north of the Azores and hits the British Isles as a Force 8 or stronger gale, having maintained a recognizable inner core.

Exposure to upper-level westerlies. Frequently, the ridge of high pressure that keeps the hurricane heading toward the west erodes. In the Atlantic Ocean, this may result in strong upper-level westerlies diving down and impinging on the northern edge of the hurricane's upper-level support structure. Poleward flow is restricted. A hurricane can still maintain itself when outflow is restricted in one quadrant. If outflow becomes too restricted, resulting shear can disrupted the inner core.

After landfall. Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts well exceeding 200 mph—which roared up the Mississippi Valley and eventually exited off the East Coast. Camille maintained tropical storm strength as far as Memphis, Tennessee. Most hurricanes will diminish in strength rapidly after landfall, reaching tropical depression strength by 48 to 72 hours. The main threat from the dying storm is from tornadoes and inland flooding. The right-hand quadrant of a hurricane or typhoon (in the Northern Hemisphere, left-hand in the Southern Hemisphere) is most frequently the strong side of the storm. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the right-hand side of the hurricane contains the strong on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life, will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.

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After landfall. Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can project life-threatening hurricane force winds over two hundred miles inland. As the storm progresses inland, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The best example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts well exceeding 200 mph—which roared up the Mississippi Valley and eventually exited off the East Coast. Camille maintained tropical storm strength as far as Memphis, Tennessee. Most hurricanes will diminish in strength rapidly after landfall, reaching tropical depression strength by 48 to 72 hours. The main threat from the dying storm is from tornadoes and inland flooding. The right-hand quadrant of a hurricane or typhoon (in the Northern Hemisphere, left-hand in the Southern Hemisphere) is most frequently the strong side of the storm. This is because the forward motion is added to the counter-clockwise punch of the storm. Thus, the right-hand side of the hurricane contains the strong on-shore flow. This is where the maximum storm surge flooding, and the greatest potential loss of life, will be experienced. After landfall, the circulation's motion over land causes a great deal of turbulence, which results in tornadoes. These are especially likely in the forward right-hand side of the storm's path.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, and finally Landfall.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface over an area 200 to 300 miles across. The result is weakening. It is possible for a hurricane to stall in one area long enough that it dissipates. In 2004, Hurricane Frances stalled off the southeast Florida coast long enough that the core of the hurricane collapsed. Frances eventually made landfall as a Category 1 hurricane, instead of the earlier expected Category 3.

Entraining dry air. Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as part of their interaction with frontal troughs that carry cool, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains deeply enough, the result can be dissipation.

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